The HSTS header tells the browser it should only ever request resources for the domain in question over HTTPS. The HSTS header includes a `max-age` directive and a directive that specifies whether subdomains should be included. It also includes a `preload` directive that tells the browser you would like to add your domain to its preload list, ensuring the first request cannot be made over HTTP on subsequent visits.

NOTE The HSTS header is ignored by browsers when the request is made over HTTP because it is possible for a hacker to intercept the request and modify the header's `max-age`, for example. So you must include HTTPS redirection in your application to ensure requests are redirected to HTTPS. Once that redirection has taken place and the connection is secured via HTTPS, the browser will honor the directives within the HSTS header for all future requests. From then on the browser will replace HTTP with HTTPS when making any requests for the specified domain.

You configure HSTS middleware via the service container. In the following listing, the `max-age` is set to one year, subdomains are included, and you ask the browser to consider adding the domain to its preload list (https://hstspreload.org/).

Listing 13.1 Configuring HSTS middleware

```
builder.Services.AddHsts(options =>
{
    options.Preload = true;                    ❶
    options.IncludeSubDomains = true;          ❷
    options.MaxAge = TimeSpan.FromDays(365);   ❸
});
```

❶ Request inclusion in the browser's preload list.

❷ Include all subdomains.

❸ Set an expiry of one year from now. This is a sliding expiry, which is updated each time the browser accesses the website.

When configured to use HTTPS, the default project template only includes HSTS middleware when your application is not running in Development mode. I recommend you do not enable HSTS while in development in case you need to run the application without HTTPS at some stage.

Listing 13.2 Configuring HSTS middleware when not in Development mode

```
if (!app.Environment.IsDevelopment())      ❶
{
    app.UseExceptionHandler("/Error");
    app.UseHsts();                         ❷
}
```

❶ Checks to establish the mode the application is running in